

# Mars in the Making Supplement Dispenser

Team 7

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# Introduction

## Problem:

- Harsh UV/Cosmic Radiation
- Mars radiation is 2,000 millirems per day
- Humans experience 620 millirems a year

## Who?:

- Customer, NASA
- Stakeholder, Congress & Colonists
- Congress is number 1 investor!

POV: Astronauts need to be protected from the harmful radiation because it will allow for a safer and longer-lasting colonization.

# Design

## Design Requirements:

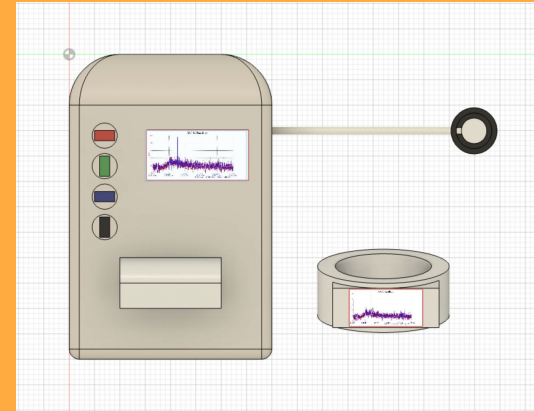
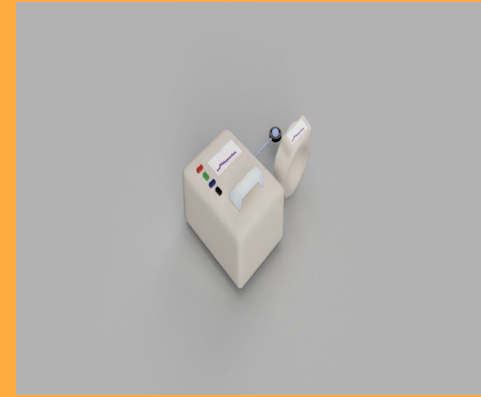
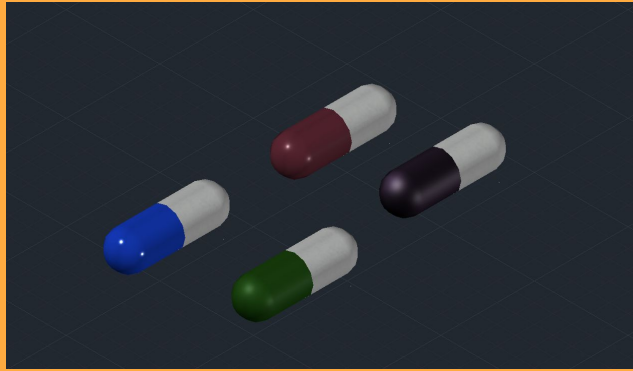
- Automated
- 2 inputs/outputs
- Address radiation concerns
- Less than 300,000 lbs
- Has moving parts

## Design Criteria:

- Low cost
- **Low weight to size ratio**
- Easy to learn
- Easy to maintain and install
- Corrosion resistant
- Strong pressure homeostasis

# Proposed design

- Sleek watch design
  - External dispenser
  - 4 Different treatments
- 4 Light sensing and dial inputs
- 4 different output actuators
- Prototype budget of \$31.50



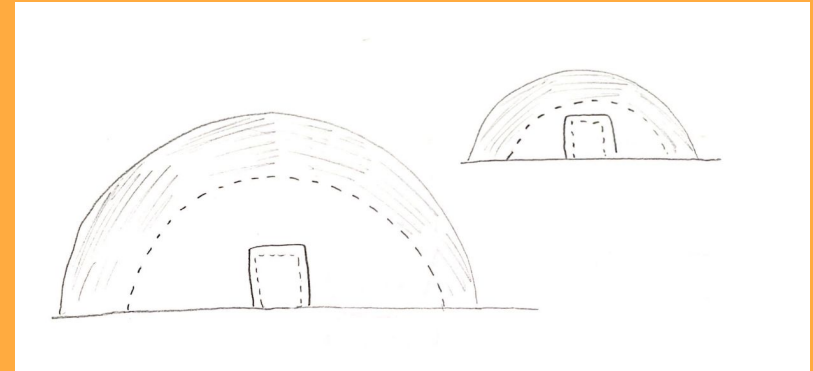
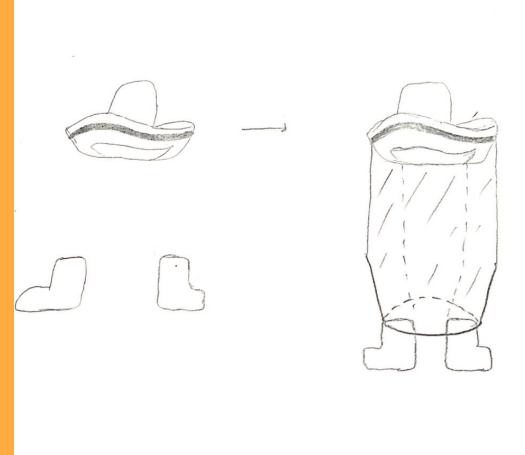
| Item                        | Amount    | Price   |
|-----------------------------|-----------|---------|
| Uno Microcontroller Board   | 1         | \$14.00 |
| Colored LED                 | 1         | \$0.10  |
| RGB Light Sensor - ISL29125 | 1         | \$5.00  |
| Small DC Motor              | 4         | \$8.00  |
| Cardboard                   | 4 sq feet | \$0.40  |
| 8x11" Construction Paper    | 4 sheets  | \$0.40  |
| Breadboard                  | 1         | \$2.00  |
| Gear attachments            | 4         | \$0.40  |
| 10-tooth Gears              | 4         | \$1.20  |

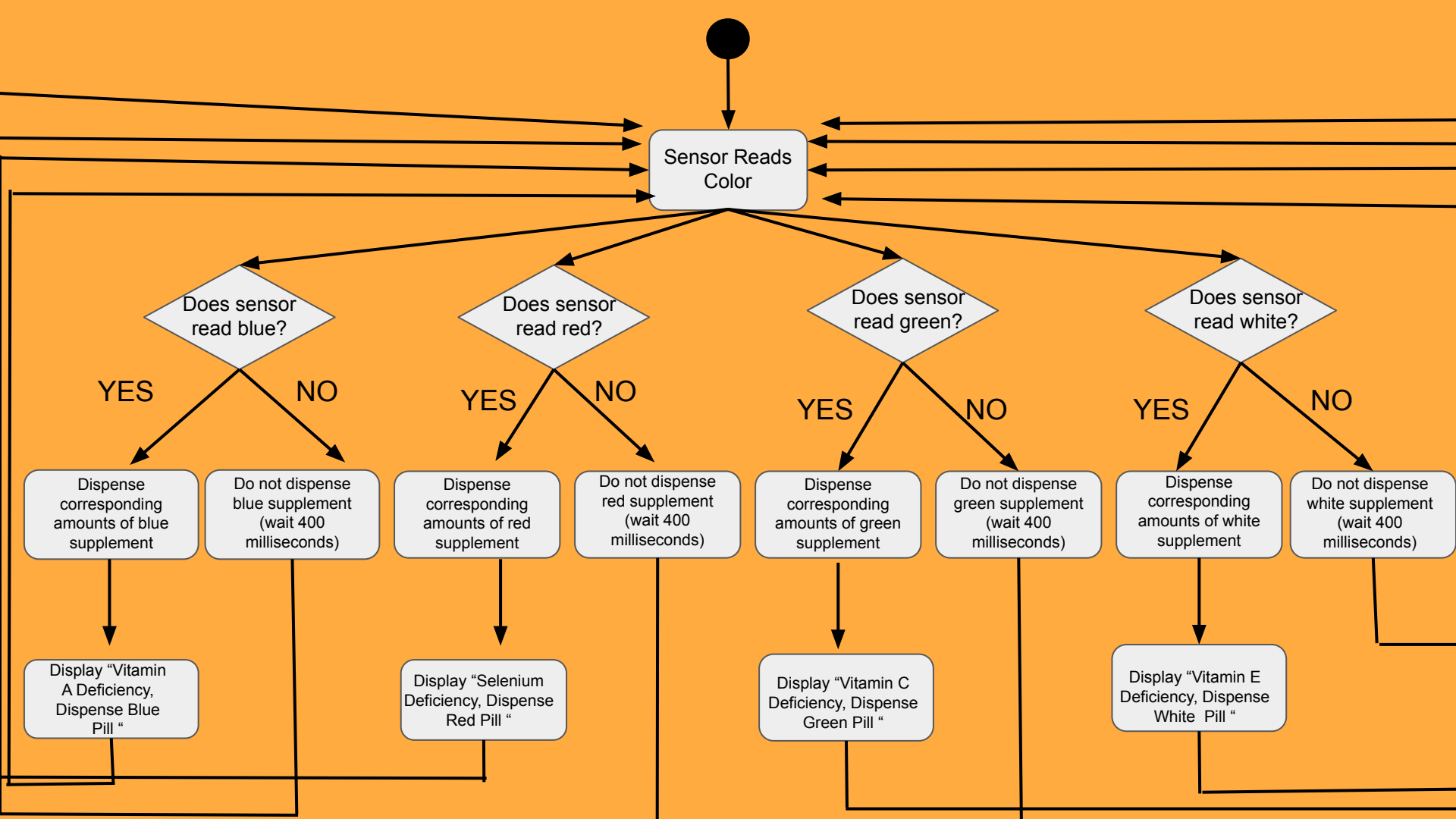
# Justification

Decision Matrix:

| Criteria                     | Weight | Design 1<br>Rating<br>Factor<br>(UV<br>resistant<br>undergarment) | Design 1<br>Weighted<br>R.F. | Design 2<br>Rating<br>Factor<br>(Personal<br>radiation<br>suit) | Design 2<br>Weighted<br>R.F. | Design 3<br>Rating<br>Factor(A<br>utomated<br>Radiation<br>Detection<br>and<br>Vitamin<br>Dispenser) | Design 3<br>Weighted<br>R.F. | Design 4<br>Rating<br>Factor<br>(Habitat<br>for a<br>Habitat) | Design 4<br>Weighted<br>R.F. |
|------------------------------|--------|-------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------------------|------------------------------|
| Low Cost                     | 0.149  | 2                                                                 | .298                         | 5                                                               | 0.745                        | 4                                                                                                    | 0.596                        | 1                                                             | 0.149                        |
| Low weight to size ratio     | 0.255  | 5                                                                 | 1.275                        | 4                                                               | 1.02                         | 5                                                                                                    | 1.275                        | 2                                                             | 0.51                         |
| Easy to learn how to operate | 0.073  | 5                                                                 | 0.365                        | 5                                                               | 0.365                        | 5                                                                                                    | 0.365                        | 3                                                             | 0.219                        |
| Easy to maintain/install     | 0.0744 | 3                                                                 | 0.2232                       | 4                                                               | 0.292                        | 5                                                                                                    | 0.372                        | 1                                                             | 0.0744                       |
| Corrosion resistant          | 0.15   | 5                                                                 | 0.75                         | 5                                                               | 0.75                         | 3                                                                                                    | 0.45                         | 5                                                             | 0.75                         |
| Strong pressure homeostasis  | 0.094  | 4                                                                 | 0.376                        | 2                                                               | 0.188                        | 1                                                                                                    | 0.094                        | 5                                                             | 0.47                         |
| Durability                   | 0.203  | 3                                                                 | 0.609                        | 2                                                               | 0.406                        | 4                                                                                                    | 0.812                        | 4                                                             | 0.812                        |
| <b>Total</b>                 | 1      |                                                                   | 3.8962                       |                                                                 | 3.772                        |                                                                                                      | 3.964                        |                                                               | 2.9844                       |

## Other Considered Designs:





# Conclusion

- Humans require vitamins
- Dispenses all necessary supplements
- NASA and Congress need this
- Automated, running 24/7

# References

- E;, Fernández-García. “Skin Protection against UV Light by Dietary Antioxidants.” *Food & Function*, U.S. National Library of Medicine, <https://pubmed.ncbi.nlm.nih.gov/24964816/#:~:text=A%20combination%20of%20vitamins%20E,solar%20UV%20radiation%20in%20humans.>
- Garner, Rob. “How to Protect Astronauts from Space Radiation on Mars.” *NASA*, NASA, 30 Sept. 2015, <https://www.nasa.gov/feature/goddard/real-martians-how-to-protect-astronauts-from-space-radiation-on-mars.>
- Jackson, Shanessa. “Life Support Systems.” *NASA*, NASA, 13 July 2016, <https://www.nasa.gov/content/life-support-systems.>



Questions?